A clear gap exists for Malagasy freshwater fish data infrastructure

No single platform currently integrates species-level data, ex situ population tracking, and cross-sector coordination for Madagascar's ~79 endemic freshwater fish species — one of the world's most threatened freshwater faunas. The data landscape is fragmented across at least a dozen disconnected systems: static PDFs, proprietary zoo databases, informal Facebook groups, and outdated IUCN assessments. This fragmentation is not merely inconvenient — it is explicitly identified in peer-reviewed literature as a barrier to saving species from extinction. Building a centralized, open-source repository with both public species information and restricted coordination tools would fill a genuine, documented gap that no existing or in-development platform adequately addresses.

The stakes are high. **Roughly 63% of Madagascar's endemic freshwater fish are threatened with extinction**, and 31 of 50 threatened species — including 11 Critically Endangered — have zero ex situ safety-net populations anywhere on Earth. The information needed to change this exists, but it is scattered across incompatible systems that don't talk to each other.

What already exists: a patchwork of partial solutions

The current data landscape for Malagasy freshwater fish can be mapped across two axes — species-level biological data and coordination/breeding management tools — with significant gaps in both.

For species-level data, the IUCN Red List provides conservation assessments for most described Malagasy freshwater fish species, but roughly a quarter remain Data Deficient, and many assessments are outdated. Ziegler et al. (2020) recommended upgrading several species (e.g., *Pachypanchax sakaramyi* from Endangered to Critically Endangered), but formal reassessments lag years behind field reality. FishBase offers the broadest per-species ecological data — morphology, distribution, ecology, genetics — but as a global generalist database, its Malagasy endemic coverage is thin and uneven. Neither platform provides Madagascar-specific conservation action tracking, population monitoring, or ex situ information.

The most comprehensive single source is the Leiss et al. (2022) *Zoo Biology* paper, which compiled a table of all 173 fish species in Malagasy freshwater habitats with IUCN status, endemism classification, population trends, and ex situ records from both ZIMS and Zootierliste. But this is a static peer-reviewed publication, not a living database — it captured a 2020 snapshot that is already outdated. Madagascar's GBIF node (MadBIF) and the REBIOMA biodiversity portal provide occurrence records but have sparse freshwater fish coverage and suffer from institutional data-sharing challenges. The Fricke et al. (2018) annotated checklist of Madagascar's fishes — at **432 pages** — is the most taxonomically complete reference but is a static PDF covering all marine and freshwater fishes without conservation-specific data.

SHOAL, the most prominent global freshwater fish conservation partnership, launched its ambitious 1,000 Fishes Blueprint in April 2024, identifying 2,300+ priority species globally. Madagascar species are included (e.g., *Typhleotris pauliani*). However, SHOAL's website is campaign- and advocacy-oriented with **no publicly accessible species-level database, no searchable registry, and no interactive data tools**. The Blueprint itself is a PDF document, not a queryable platform. Similarly, the IUCN SSC Freshwater Fish Specialist Group (FFSG) — the Red List Authority for freshwater fish with 160+ expert members — maintains a website with newsletters and species spotlights but **no species database, no conservation status dashboard, and no coordination tools**.

The bottom line: species-level data exists but is scattered across the IUCN Red List, FishBase, published papers, and institutional databases with no Madagascar-specific integration point.

Coordination tools are siloed between professional and hobbyist sectors

The coordination landscape reveals a striking divide: **professional zoo networks use sophisticated but proprietary tools that exclude hobbyists and researchers, while hobbyist networks rely on informal systems that lack scientific rigor**.

ZIMS (Species360), the gold standard for captive animal records, tracks 22,000+ species across 1,300+ member institutions with real-time inventory, medical records, and studbook management. It was the primary data source for the Leiss et al. (2022) audit, which identified **21 Malagasy freshwater fish species held across global zoos totaling ~4,267 individuals**. ZIMS integrates with the Species Conservation Toolkit Initiative (SCTI), providing population viability analysis (Vortex) and genetic management (PMx) tools. But ZIMS is proprietary, membership-fee-based, and restricted exclusively to accredited institutions. Private hobbyists and independent researchers cannot access or contribute data — a critical limitation given that hobbyist breeders hold at least 3 additional Malagasy species not found in any zoo.

The AZA Freshwater Fishes TAG and the **EAZA/EUAC Freshwater Teleost TAG** both manage formal breeding programs (SSPs and EEPs respectively) for Malagasy species. The EAZA TAG operates dedicated EEPs for Bedotiidae, Cichlidae, and Aplocheilidae — all families containing Malagasy endemics. Charles-Édouard Fusari (Aquarium Tropical du Palais de la Porte Dorée, Paris) coordinates the Bedotiidae EEP and Fish Net Madagascar. European institutions hold the majority of captive Malagasy freshwater fish (**3,602 individuals** versus 659 in North America). But all TAG coordination tools — Regional Collection Plans, Breeding and Transfer Plans, studbooks — are restricted to member institutions with no public-facing components.

On the hobbyist side, the **CARES Preservation Program** (American Cichlid Association) maintains publicly viewable priority species lists with strong Madagascar coverage — Dr. Paul V. Loiselle coordinates the Madagascar region, and Jim Cumming serves as Madagascan Cichlid Species Specialist. CARES includes Malagasy Bedotiidae and Cichlidae on its priority lists. However, the program runs on a basic WordPress site with **static HTML species lists, no dynamic database, no real-time inventory tracking, no genetic management tools, and paper-based registration through member clubs**. Coordination happens through personal contacts, Facebook groups ("Madagascar Endangered Fishes," "CARES Preservation"), and conference presentations.

Citizen Conservation (Germany), co-founded by Frogs & Friends and Cologne Zoo, represents the most promising bridge between these sectors. It runs structured breeding programs where CC owns the animals and distributes them to vetted private breeders who report inventory twice yearly. CC currently breeds several Malagasy species, including *Bedotia madagascariensis*, *Ptychochromis loisellei*, and *Ptychochromis insolitus* (the "rarest fish in the world"). Critically, CC is developing **"Wild at Home"** — described by co-founder Björn Encke as "a cross between ZIMS and Facebook for wildlife keepers." Funded by the German Federal Ministry for Economic Affairs, it features inventory dashboards, animal group management, and community networking. However, Wild at Home remains in **password-protected beta**, is not yet publicly available, and its timeline for broader release is unclear.

The Goodeid Working Group offers a useful organizational model — with regional coordinators, species-specific population coordinators, and partnerships linking ex situ (hobbyists/zoos) to in situ (Fish Ark Mexico) efforts — but covers only Mexican livebearers with no Malagasy relevance.

The literature explicitly documents fragmentation as a barrier

Multiple peer-reviewed publications and institutional statements identify information fragmentation as a direct obstacle to Malagasy freshwater fish conservation.

Leiss et al. (2022) — the landmark "call for action" paper — had to triangulate data from three disconnected sources (ZIMS, Zootierliste, and a Facebook group questionnaire) just to compile an inventory of captive populations, explicitly noting that "the ZIMS list may miss some captive populations or breeding records." The paper calls for establishing studbooks and monitoring programs, better distribution of offspring including to private breeders via citizen conservation, closer connection between field research and breeding efforts, and **development of effective, integrated in situ and ex situ conservation strategies** under the IUCN One Plan Approach. The fact that no formal studbooks exist for most Malagasy freshwater fish species — despite some having been in captivity for decades — underscores the coordination deficit.

Fusari himself articulated the problem when his institution joined Species360: "In terms of conservation, it will be great to have a more global view on which threatened species are held in which institution and in which numbers. I think it will also facilitate the coordination of zoological institutions for ex situ conservation efforts of highly threatened species." This statement implicitly acknowledges that such a global view does not currently exist in an accessible form.

Additional evidence of fragmentation includes Vences et al. (2022), which showed that even **basic genetic identification tools (DNA barcodes) are only now being developed** for Malagasy freshwater fish — making taxonomic confusion in captive holdings a real problem. Ravelomanana, Fusari et al. (2024) demonstrated that species assumed extinct are being rediscovered using eDNA surveys, indicating that field monitoring data is severely incomplete. And the Máiz-Tomé et al. (2018) IUCN regional assessment found 43% of Madagascar's freshwater species threatened with nearly a quarter Data Deficient, concluding that "our knowledge of Madagascar's unique freshwater species remains incomplete."

Notably, **no formal, comprehensive conservation action plan for Malagasy freshwater fishes has ever been published**. Reinthal and Stiassny called for a conservation strategy in 1991 — 35 years ago — and one still does not exist. The SHOAL Blueprint is the closest approximation but is global in scope, not Madagascar-specific, and is a static PDF rather than a living coordination tool.

Comparable models show what a purpose-built platform could look like

Three existing systems offer partial architectural models, though none fully addresses the combined needs of Malagasy freshwater fish conservation.

Amphibian Ark (AArk) provides the strongest organizational model. Its Conservation Needs Assessment database has evaluated 2,300+ amphibian species through structured 26-question assessments, with results publicly viewable online. Its Program Progress Database tracks ex situ programs with filterable data. AArk delegates animal-level record-keeping to ZIMS while maintaining its own species-level coordination layer — an elegant separation of concerns. However, AArk lacks a formal restricted-access backend for breeding coordination; that function operates through personal relationships and institutional partnerships rather than digital infrastructure.

BirdLife International's DataZone, relaunched in 2025, demonstrates the hybrid public/restricted data model most clearly. It provides 11,000+ public species factsheets alongside restricted spatial data requiring application-based access for non-commercial use. BirdLife's partnership in IBAT (Integrated Biodiversity Assessment Tool) shows how conservation data can be structured with tiered access — free for conservation purposes, subscription-based for commercial screening.

ZIMS/Species360 remains the technical gold standard for animal-level records at scale, with proven role-based access architecture serving 1,300+ institutions. But its proprietary, membership-fee model and

exclusion of non-institutional participants make it unsuitable as a coordination platform for the broader Malagasy freshwater fish community. The key insight from ZIMS is that **any new platform should complement rather than duplicate it** — focusing on species-level coordination, public education, field program tracking, and cross-sector networking that ZIMS cannot provide.

On the technical side, **Darwin Core** is the clear base standard for biodiversity data interoperability, used by GBIF (hundreds of millions of records) and compatible with all major open-source platforms. Conservation breeding–specific data (genetics, pedigree, husbandry protocols) would require custom extensions beyond standard Darwin Core. For role-based access, **GBIF's sensitive species best practices** provide established protocols for coordinate generalization across sensitivity tiers, and **Symbiota's five-tier access model** (public → editor → collection manager → rare species reader → administrator) offers a proven template.

Several mature open-source platforms could serve as foundations. **Symbiota** (PHP/MySQL, Darwin Core–native, 2,000+ collections, 95 million records) and **Specify** (Java + Python/JavaScript, 70+ consortium members) are battle-tested but designed for museum specimens rather than conservation breeding. **OpenBioMaps** (PostgreSQL-PostGIS, self-hosted) handles both public and sensitive data with flexible role-based access and is the most adaptable to a hybrid public/restricted model. The emerging **BiOS** platform (2026) offers modern architecture with a PostgreSQL-PostGIS backend, RESTful API, and decoupled frontend specifically designed for scalability and interoperability. **No existing open-source platform is purpose-built for conservation breeding coordination** — this remains the clearest technical gap in the ecosystem.

The verdict: build it, but build it strategically

The evidence strongly supports creating a new centralized platform, but with important caveats about scope and positioning.

The case for building is compelling. There is no existing or in-development platform that combines: (a) public-facing Malagasy freshwater fish species profiles integrating IUCN, FishBase, and field data; (b) restricted coordination tools for tracking ex situ populations across zoos, aquariums, and private breeders; (c) field program monitoring linking in situ surveys to ex situ management; and (d) cross-sector networking between professional institutions and hobbyist communities. This gap is not a matter of poor visibility — it reflects a genuine absence of infrastructure. The closest emerging solution, Citizen Conservation's Wild at Home, is in early beta, is not freshwater fish–specific, and may not achieve the breadth of public-facing species information or researcher-facing data tools that the community needs.

However, the platform should be designed to link to existing systems rather than replace them. Specifically:

- **Complement ZIMS** by providing the species-level coordination layer, public information, and cross-sector networking that ZIMS cannot offer — while drawing on ZIMS data (via Species360 Insights or institutional data-sharing agreements) for captive population snapshots
- **Integrate with IUCN Red List and FishBase** data through APIs rather than duplicating their species assessments
- **Publish occurrence data to GBIF** via Darwin Core Archives and the Integrated Publishing Toolkit
- **Support CARES and Citizen Conservation** by providing digital infrastructure that upgrades their currently manual coordination processes
- **Align with SHOAL and FFSG** priorities by operationalizing the 1,000 Fishes Blueprint for Malagasy species specifically

A recommended architecture would use a **PostgreSQL-PostGIS backend with RESTful API and a decoupled modern web frontend**, built on Darwin Core with custom extensions for breeding coordination

data. Five access tiers — public, registered researcher, conservation coordinator, program manager, and administrator — would follow the Symbiota/BirdLife model. The public layer would serve as the "single source of truth" for Malagasy freshwater fish species profiles, while restricted layers would handle sensitive location data, captive population genetics, breeding recommendations, and transfer coordination.

The strongest argument for acting now is that **the community is small enough to coordinate but the crisis is too urgent to wait**. With fewer than 80 endemic species, a manageable number of institutional and hobbyist stakeholders, and key figures like Fusari, Ziegler, and Loiselle already in active communication, a purpose-built platform could achieve comprehensive coverage relatively quickly. The 2022 Leiss et al. paper provides a ready-made baseline dataset. Citizen Conservation's Wild at Home development could potentially be leveraged or integrated rather than built from scratch. And the absence of a published conservation action plan for this fauna means the platform itself could catalyze the strategic planning process the community has needed for decades.

The risk of not building is that coordination continues to depend on personal relationships, conference hallway conversations, and Facebook groups — systems that don't scale, don't persist when key individuals move on, and can't support the systematic genetic and demographic management that these species desperately need.